Lecture 6

CHAPTER 3

Variational approach to stochastic partial differential equations

3.1. Insertion from functional analysis: Weak topology

We start this chapter with a short insertion on basic functional analytic facts needed in the following. Basic facts concerning weak topology, Banach-Alaoglu's theorem, Riesz' representation theorem will be needed. In particular we will need the basic fact that weak convergece + norm convergence implies strong convergence in Hilbert spaces.

3.2. Variational approach to partial differential equations

Having recalled the required basic functional analytic facts, we are now ready for the introduction to the variational approach to stochastic partial differential equations. We first explain the basic concept in the deterministic case. To this end we will first need the concept of a Gelfand triple.

Let H be a separable, real Hilbert space and let $V \subseteq H$ be a second Hilbert space such that the embedding $V \hookrightarrow H$ is dense and continuous. We assume that

$$\|h\|_H \leq \|h\|_V$$

for all $h \in V$.

Identifying H with its dual space $H' = L(H, \mathbb{R})$ (using the Riesz-Isomorphism) we obtain a dense and continuous embedding

$$H \cong H' \hookrightarrow V'$$

via

$$\underbrace{h}_{\in H \cong H'} \quad \underbrace{\cong}_{\substack{\text{is the}\\ \text{embedding}}} \quad \underbrace{(v \mapsto \langle h, v\rangle_H, v \in V)}_{\in L(V,\mathbb{R}) = V'}$$

for all $h \in H$. This means, that an element h from $H \cong H'$ is identified with the linear functional

$$l_h(v) := \langle h, v\rangle_H, \quad v \in V.$$

Note that

$$\|l_h\|_{V'} := \sup_{\|v\|_V \leq 1} \underbrace{l_h(v)}_{\langle h,v\rangle_H} \leq \sup_{\|v\|_V \leq 1} \|h\|_H \|v\|_H \leq \sup_{\|v\|_V \leq 1} \|h\|_H \|v\|_V \leq \|h\|_H,$$

hence, indeed, the embedding $H \equiv H' \hookrightarrow V'$ is continuous.

Let $A \in L(V, V')$ be a bounded (hence continuous) linear operator $A : V \longrightarrow V'$, satisfying the following **coercivity condition**

$\exists \lambda, \alpha > 0$ such that

$$-\langle Au, u\rangle \ \geq \alpha \|u\|_V^2 - \lambda \|u\|_H^2 \quad \forall u \in V.$$

EXAMPLE 3.1. Let $D \subset \mathbb{R}^d$ be an open subset, $H = L^2(D), V = H^1(D) := H^{1,2}(D)$, where

$$H^1(D) = \{u \in L^2(D) : \partial_{x_i} u \in L^2(D), i = 1, ..., d\}$$

is the Sobolev space of one times weakly differentiable functions f with f and all partial derivatives in $L^2(D)$.

Remark (weak derivatives - Sobolev spaces) $\partial_{x_i} u$ is understood in the weak sense, i.e., $u \in H^1(D)$ if and only if for all i there exists $u^{(i)} \in L^2(D)$ such that the following integration by parts formula

$$\int_D \partial_{x_i} \varphi(x) u(x)\, dx = -\int_D \varphi(x) u^{(i)}(x)\, dx \tag{3.1}$$

holds for all "test-functions" $\varphi \in C_c^1(D)$. Here, $C_c^1(D)$ denotes the space of one times continuously differentiable functions having compact support in D. The function $u^{(i)}$ is uniquely determined via the integration by parts formula (3.1). Indeed, let $\tilde{u}^{(i)}$ be another weak derivative, satisfying the integration by parts formula (3.1), then

$$\int_D \varphi(x) \left(u^{(i)}(x) - \tilde{u}^{(i)}(x)\right) dx = -\int_D \partial_{x_i} \varphi(x) u(x) dx + \int_D \partial_{x_i} \varphi(x) u(x) dx = 0$$

for all $\varphi \in C_c^1(D)$ and thus $u^{(i)} = \tilde{u}^{(i)}$.

It is clear that for $u \in C_c^1(\mathbb{R}^d)$ the weak derivative $\partial_{x_i} u$ (in $L^2(\mathbb{R}^d)$) exists and coincides with the usual partial derivative, since

$$\int_D \partial_{x_i} \varphi(x) u(x)\, dx = -\int_D \varphi(x) \partial_{x_i} u(x)\, dx$$

$H^1(D) = H^{1,2}(D)$ is again a separable real Hilbert space w.r.t. the inner product

$$\langle u, v \rangle := \int_D uv dx + \sum_{i=1}^d \int_D \partial_{x_i} u \partial_{x_i} v dx.$$

Denote with $H_0^1(D) \subset H^1(D)$ the closure of $C_c^1(D) \subset H^1(D)$. Then this closure is called Sobolev space (of order 1) with Dirichlet boundary conditions.

Illustration ($d = 1$). Let $D = (0, 1)$. $u \in H^1(D)$ implies that u is absolute continuous, i.e.,

$$u(x) - u(y) = \int_x^y u'(s) ds$$

with Radon-Nikodym derivative $u' \in L^2(D)$. In particular, u has a continuous version $\tilde{u}$. Moreover, $u \in H_0^1(D)$ if and only if $u \in H^1(D)$ and $\tilde{u}(0) = \tilde{u}(1) = 0$.

Having introduced the Gelfand triple

$$H^1(D) \hookrightarrow L^2(D) = L^2(D)' \hookrightarrow H^1(D)'$$

we can now realize the Laplace operator

$$\Delta := \sum_{i=1}^{d} \partial^2_{x_i x_i}$$

as a continuous (!) linear operator $\Delta : H^1(D) \longrightarrow H^1(D)'$ as follows:
Given some $u \in H^1(D)$ let $\Delta u \in H^1(D)'$ be the unique linear functional defined by

$$v \mapsto {}_{H^1(D)'}\langle \Delta u, v\rangle_{H^1(D)} = -\sum_{i=1}^{d} \int_D \partial_{x_i} u \, \partial_{x_i} v \, dx, \quad v \in H^1(D).$$

Let us now consider the initial value problem

$$\begin{cases} \frac{du}{dt}(t) &= Au(t) + f(t),\ t \geq 0 \\ u(0) &= u_0. \end{cases} \tag{3.2}$$

THEOREM 3.2. *Let $A \in L(V, V')$ be coercive, $u_0 \in H$ and $f \in L^2([0,T]; V')$. Then equation* (3.2) *has a unique solution*

$$u \in C([0,T]; H) \cap L^2([0,T]; V).$$

We need the following lemma for the proof:

LEMMA 3.3. *Let $u \in L^2([0,T]; V)$ be such that $t \mapsto u(t)$ is absolute continuous with values in V' and suppose that $\frac{du}{dt} \in L^2([0,T]; V')$. Then $u \in C([0,T]; H)$ and*

$$\frac{1}{2}\frac{d}{dt}\|u(t)\|^2_H = {}_{V'}\langle \frac{du}{dt}(t), u(t)\rangle_V \quad a.e..$$

PROOF. **Step 1** Let $u_m \in C^1([0,T]; V)$, $m \geq 1$, such that $\lim_{m\to\infty} \|u_m - u\|_{L^2([0,T];V)} + \|u_m' - u'\|_{L^2([0,T];V')} = 0$. Then

$$\lim_{m\to\infty} \frac{d}{dt}\|u_m(t)\|^2_H = \lim_{m\to\infty} 2_{V'}\langle u_m'(t), u_m(t)\rangle_V = 2_{V'}\langle u'(t), u(t)\rangle_V$$

in $L^1([0,T])$ (Lebegue's convergence theorem). Since also $\lim_{m\to\infty} \|u_m(t)\|^2_H = \|u(t)\|^2_H$ in $L^1([0,T])$, it follows that $t \mapsto \|u(t)\|^2_H$ is absolute continuous with derivative $\frac{d}{dt}\|u(t)\|^2_H = 2_{V'}\langle u'(t), u(t)\rangle_V$.

Step 2 Define $\langle v\rangle := \frac{1}{T}\int_0^T v(t)\,dt$ for all $v \in L^2([0,T]; H)$. Then for $v \in C^1([0,T]; V)$

$$\begin{aligned} \|v(t)\|^2_H &\leq 2\|v(t) - \langle v\rangle\|^2_H + 2\|\langle v\rangle\|^2_H \\ &\leq \frac{2}{T}\int_0^T \|v(t) - v(r)\|^2_H \, dr + \frac{2}{T}\int_0^T \|v(r)\|^2_H \, dr. \end{aligned}$$

Using

$$\begin{aligned} \|v(t) - v(r)\|^2_H &= 2\int_r^t {}_{V'}\langle v'(s), v(s) - v(r)\rangle_V \, ds \\ &\leq \int_0^T \|v'(s)\|^2_{V'} \, ds + \int_0^T \|v(s) - v(r)\|^2_V \, ds \end{aligned}$$

we have that

$$\|v(t)\|_H^2 \leq 2\int_0^T \|v'(s)\|_{V'}^2\,ds + \frac{2}{T}\int_0^T\int_0^T \|v(s)-v(r)\|_V^2\,ds\,dr + \frac{2}{T}\int_0^T \|v(r)\|_H^2\,dr.$$

Substituting $v = u_n - u_m$, where (u_n) is as in Step 1 we see that u_n is a Cauchy sequence in $C([0,T];H)$ and its pointwise limit $\tilde{u}(t) := \lim_{n\to\infty} u_n(t)$ is again in $C([0,T];H)$ and a version of $u \in L^2([0,T];V)$. □

Proof of Theorem 3.2. **Uniqueness** Let $u, v \in L^2([0,T];V)$ be two solutions of (3.2). Then $u - v$ solves the parabolic equation

$$\frac{d(u-v)}{dt}(t) = A(u(t)-v(t))$$

with initial condition $u(0) - v(0) = 0$. Lemma 3.3 now implies that

$$\frac{1}{2}\frac{d}{dt}\|u(t)-v(t)\|_H^2 = {}_{V'}\langle A(u(t)-v(t)), u(t)-v(t)\rangle_V \overset{\text{coercive}}{\leq} \lambda\|u(t)-v(t)\|_H^2.$$

Consequently,

$$\begin{aligned}\frac{d}{dt}\left(e^{-2\lambda t}\|u(t)-v(t)\|_H^2\right) &= e^{-2\lambda t}\left(-2\lambda\|u(t)-v(t)\|_H^2 + 2\langle A(u(t)-v(t)), u(t)-v(t)\rangle\right)\\ &\leq e^{-2\lambda t}\left(-2\lambda\|u(t)-v(t)\|_H^2 + 2\lambda\|u(t)-v(t)\|_H^2\right) = 0,\end{aligned}$$

which implies $e^{-2\lambda t}\|u(t)-v(t)\|_H^2 \leq \|u(0)-v(0)\|_H^2 = 0$, thus

$$\|u(t)-v(t)\|_H = 0$$

and therefore $u(t) - v(t) = 0$, proving the uniqueness part.

Existence
We will apply an approximation procedure called **Galerkin-method**. To this end let $(e_k)_k$ be a CONS of H consisting of elements in V. It is always possible to choose such a system, since V is dense in H. Let

$$V_n := \operatorname{span}\{e_1, ..., e_n\}$$

be the n-dimensional subspace of V (and H too) spanned by the first n basis vectors. It follows from the theory of (finite) systems of linear ordinary differential equations that for all $n \in \mathbb{N}$ there exists $u_n \in C([0,T];V_n)$ such that

$$\begin{cases}\frac{d}{dt}\langle u_n(t), e_k\rangle &= \langle Au_n(t), e_k\rangle + \langle f(t), e_k\rangle,\ t \geq 0\\ \langle u_n(0), e_k\rangle &= \langle u_0, e_k\rangle\end{cases} \tag{3.3}$$

for $k = 1, \ldots, n$.

The next step consist of taking the limit $n \to \infty$. An essential ingredient for this is to derive the following a priori estimate

$$\sup_{n\geq 1}\left(\sup_{t\in[0,T]}\|u_n(t)\|_H^2 + \int_0^T \|u_n(t)\|_V^2\,dt\right) < \infty \tag{3.4}$$

which implies that the approximating sequence $(u_n)_{n\geq 1}$ will be bounded in the appropriate function spaces. For the proof of the a priori estimate note that (3.3) implies

$$\begin{aligned}
\frac{1}{2}\frac{d}{dt}\|u_n(t)\|_H^2 &= \frac{1}{2}\frac{d}{dt}\sum_{k=1}^{n}\langle u_n(t), e_k\rangle^2 \\
&= \sum_{k=1}^{n}\langle Au_n(t) + f(t), e_k\rangle\langle u_n(t), e_k\rangle \\
&= \langle Au_n(t) + f(t), u_n(t)\rangle \\
&\leq -\alpha\|u_n(t)\|_V^2 + \lambda\|u_n(t)\|_H^2 + \|f(t)\|_{V'}\|u_n(t)\|_V \\
&\leq -\alpha\|u_n(t)\|_V^2 + \lambda\|u_n(t)\|_H^2 + \frac{1}{\sqrt{\alpha}}\|f(t)\|_{V'}\sqrt{\alpha}\|u_n(t)\|_V \\
&\underset{\text{ineq.}}{\overset{\text{Young-}}{\leq}} -\frac{\alpha}{2}\|u_n(t)\|_V^2 + \lambda\|u_n(t)\|_H^2 + \frac{1}{2\alpha}\|f(t)\|_{V'}^2.
\end{aligned}$$

Integrating both sides of the inequality w.r.t. t yields that

$$\begin{aligned}
&\|u_n(t)\|_H^2 + \alpha\int_0^t \|u_n(s)\|_V^2 ds \\
&\qquad \leq \|u_0\|_H^2 + 2\lambda\int_0^t \|u_n(s)\|_H^2 ds + \frac{1}{\alpha}\int_0^t \|f(s)\|_{V'}^2 ds \\
&\qquad \leq \|u_0\|_H^2 + \frac{1}{\alpha}\int_0^t \|f(s)\|_{V'}^2 ds + 2\lambda\int_0^t \left(\|u_n(s)\|_H^2 + \alpha\int_0^s \|u_n(r)\|_V^2 dr\right) ds
\end{aligned}$$

Finally, Gronwall's lemma implies the a priori estimate (3.4).
Using this boundedness, we can extract a subsequence, again denoted with u_n, converging weakly to some $u \in L^2([0,T];V)$. Since $Au_n \longrightarrow Au$ weakly in $L^2([0,T];V')$ we obtain from (3.3) that

$$\frac{d}{dt}\langle u(t), e_k\rangle = \langle Au(t) + f(t), e_k\rangle, \qquad k = 1, 2, \ldots$$

and therefore

$$\frac{d}{dt}u(t) = Au(t) + f(t)$$

which proves the assertion. □

3.2.1. Generalization to the nonlinear case. We assume that $A : V \longrightarrow V'$ is still continuous and coercive, but not necessarily linear. The a priori estimate (3.4) in the existence part in the proof of Theorem 3.2 remains true, so that we can still extract a weakly convergent subsequence $u_n \to u$ in $L^2([0,T];V)$ satisfying $A(u_n) \to g$ in $L^2([0,T];V')$ for some unknown element g. In contrast to the linear case we cannot immediately conclude that $A(u) = g$. In order to do this, there are the following two strategies:

1. Monotonicity (Dissipativity)

We assume the following monotonicity condition

$$\langle A(u) - A(v), u - v\rangle \leq \lambda \|u - v\|_H^2 \quad \text{for some } \lambda \in \mathbb{R}, \tag{3.5}$$

together with the boundedness assumption of the type

$$\|A(u)\|_{V'} \leq M(1 + \|u\|_V),$$

and a certain continuity assumption (see below).
For simplicity assume that (3.5) holds with $\lambda = 0$, otherwise we can pass w.l.o.g. from the operator $A(\cdot)$ to the operator $A(\cdot) - \lambda I$. Then

$$\int_0^T \langle A(u_n) - A(v), u_n - v\rangle(t)dt \leq 0 \quad \forall v \in L^2([0,T];V).$$

The left hand side can be divided up into four terms. Three of these terms,

$$\int_0^T \langle A(u_n), v\rangle(t)dt, \ \int_0^T \langle A(v), u_n\rangle(t)dt \text{ and } \int_0^T \langle A(v), v\rangle(t)dt$$

converge towards the expected limit. The only remaining difficult term is

$$\int_0^T \langle A(u_n), u_n\rangle(t)dt = \frac{1}{2}\left(\|u_n(T)\|_H^2 - \sum_{k=1}^n \langle u_0, e_k\rangle^2\right) - \int_0^T \langle f, u_n\rangle(t)dt.$$

Again, the second and third term converge towards the expected limit, the first term not necessarily. But, since u_n, $n \geq 1$, is bounded, we can extract a subsequence such that both, $u_n \to u$ weakly in $L^2([0,T];V)$ und $u_n(T) \to u(T)$ weakly in H. Similar to the linear case we then have

$$\frac{d}{dt}\langle u(t), e_k\rangle \ = \ \langle g(t) + f(t), e_k\rangle, \quad k = 1, 2, \ldots$$

and therefore $\frac{d}{dt}u(t) = g(t) - f(t) \in V'$ which implies, using Lemma 3.3, that

$$\begin{aligned}
\int_0^T \langle g(t), u(t)\rangle dt &= \frac{1}{2}\left(\|u(T)\|_H^2 - \|u_0\|_H^2\right) - \int_0^T \langle f(t), u(t)\rangle dt \\
&\leq \limsup_{n\to\infty} \frac{1}{2}\left(\|u_n(T)\|_H^2 - \sum_{k=1}^n \langle u_0, e_k\rangle^2\right) - \int_0^T \langle f(t), u_n(t)\rangle dt \\
&\leq \limsup_{n\to\infty} \int_0^T \langle A(u_n(t)), u_n(t)\rangle dt.
\end{aligned}$$

In fact we have for all $v \in L^2([0,T];V)$

$$\int_0^T \langle g(t) - A(v(t)), u(t) - v(t)\rangle dt \leq \limsup_{n\to\infty} \int_0^T \langle A(u_n(t)) - A(v(t)), u_n(t) - v(t)\rangle dt \leq 0.$$

Choosing $v(t) = u(t) + \theta w(t)$ for any $w \in L^2([0,T];V)$ and $\theta > 0$, implies that

$$\int_0^T \langle g(t) - A((u+\theta w)(t)), w(t)\rangle dt \leq 0.$$

Taking the limit $\theta \searrow 0$ and assuming that $\theta \mapsto A((u+\theta w)(t))$ is continuous, we arrive at

$$\int_0^T \langle g(t) - A(u(t)), w(t)\rangle dt \leq 0 \quad \forall w \in L^2([0,T];V).$$

Since w is arbitrary we can conclude from this $g = A(u)$.

This last argument is called the **"monotonicity trick"**.

2. Compactness method

The basic additional assumption in this method is that the embedding $V \hookrightarrow H$ is compact (and not only continuous). This means that that every bounded subset $A \subset V$ is precompact in H. This is not in general true for infinite dimensional spaces, since the Bolzano-Weierstrass theorem only holds in finite dimensional vector spaces.

LEMMA 3.4. *Let the embedding $V \hookrightarrow H$ be compact. Then*

$$X := L^2([0,T];V) \cap H^{1,2}([0,T];V') \hookrightarrow L^2([0,T];H)$$

is compact too. In particular, any sequence (u_n) bounded in X contains a subsequence strongly convergent in $L^2([0,T];H)$.

PROOF. For $\varepsilon > 0$ define

$$J_\varepsilon g(s) := \frac{1}{2\varepsilon}\int_{s-\varepsilon}^{s+\varepsilon} g(t)\,dt = \frac{1}{2\varepsilon}\int_{-\varepsilon}^{\varepsilon} g(s+t)\,dt.$$

(We extend any integrable mapping $g : [0,T] \to V$ by 0 to $g : \mathbb{R} \to V$.) Clearly, J_ε defines a bounded linear operator from $L^2([0,T];V)$ into $C([0,T];V)$ (resp. from $L^2([0,T];H)$ into $C([0,T];H)$), since

$$\begin{aligned}\|J_\varepsilon g(s)\|_V &= \frac{1}{2\varepsilon}\left\|\int_{-\varepsilon}^{\varepsilon} g(s+t)\,dt\right\|_V \leq \frac{1}{2\varepsilon}\int_{-\varepsilon}^{\varepsilon}\|g(s+t)\|_V\,dt \\ &\leq \frac{1}{\sqrt{2\varepsilon}}\left(\int_{-\varepsilon}^{\varepsilon}\|g(s+t)\|_V^2\,dt\right)^{\frac{1}{2}} \leq \frac{1}{\sqrt{2\varepsilon}}\|g\|_{L^2([0,T];V)}.\end{aligned}$$

We will next show that for $\varepsilon > 0$ the mapping $J_\varepsilon : X \to C([0,T];H)$ is compact using Arzela-Ascoli's theorem. To this end first note that for $g \in X$

$$\frac{d}{ds}J_\varepsilon g(s) = \underbrace{\frac{1}{2\varepsilon}(g(s+\varepsilon) - g(s-\varepsilon))}_{\in L^2([0,T];V)} = \underbrace{\frac{1}{2\varepsilon}\int_{-\varepsilon}^{\varepsilon}\frac{d}{ds}g(s+t)\,dt}_{\in L^2([0,T];V')},$$

hence

$$\begin{aligned}\|J_\varepsilon g(t_2) - J_\varepsilon g(t_1)\|_H &= \left\| \int_{t_1}^{t_2} \frac{d}{ds} J_\varepsilon g(s)\, ds \right\|_H \\ &\le \frac{1}{2\varepsilon} \int_{t_1}^{t_2} \|g(s+\varepsilon) - g(s-\varepsilon g)\|_H\, ds \\ &\le \frac{1}{2\varepsilon} \sqrt{t_2 - t_1} \left(\int_{t_1}^{t_2} \|g(s+\varepsilon) - g(s-\varepsilon g)\|_H^2\, ds \right)^{\frac{1}{2}} \\ &\le \frac{\sqrt{t_2 - t_1}}{\varepsilon} \|g\|_X.\end{aligned}$$

Suppose now that $A \subset X$ is bounded. Then $A \subset C([0,T];H)$ is bounded and uniformly equicontinuous, since

$$\sup_{|t_1 - t_2| \le \delta} \sup_{g \in A} \|g(t_1) - g(t_2)\|_H \overset{\delta \to 0}{\longrightarrow} 0.$$

Now Arzela-Ascoli's theorem can be applied to conclude that A is relatively compact in $C([0,T],H)$ and thus also in $L^2([0,T];H)$.

We will prove next that

$$\|J_\varepsilon g - g\|_{L^2([0,T];V')} \le \frac{2\sqrt{\varepsilon T}}{3} \|g\|_X. \tag{3.6}$$

This will prove the assertion, since for $g_n \in X$ converging weakly to some $g \in X$, is follows that $J_\varepsilon g_n \to J_\varepsilon g$ strongly in $L^2([0,T];H)$ for all $\varepsilon > 0$ and thus $g_n \to g$ strongly in $L^2([0,T];H)$. Indeed: given $\tilde{\varepsilon} > 0$, there exists $C_{\tilde{\varepsilon}}$ such that

$$\|u\|_{L^2([0,T];H)} \le \tilde{\varepsilon} \|u\|_{L^2([0,T];V)} + C_{\tilde{\varepsilon}} \|u\|_{L^2([0,T];V')}$$

for all $u \in L^2([0,T];V)$. Therefore,

$$\begin{aligned}&\|g_n - g\|_{L^2([0,T];H)} \\ &\quad \le \|g_n - J_\varepsilon g_n\|_{L^2([0,T];H)} + \|J_\varepsilon(g_n - g)\|_{L^2([0,T];H)} + \|J_\varepsilon g - g\|_{L^2([0,T];H)} \\ &\quad \le \tilde{\varepsilon} \|g_n - J_\varepsilon g_n\|_{L^2([0,T];V)} + C_{\tilde{\varepsilon}} \|g_n - J_\varepsilon g_n\|_{L^2([0,T];V')} + \|J_\varepsilon(g_n - g)\|_{L^2([0,T];H)} \\ &\qquad + \tilde{\varepsilon} \|J_\varepsilon g - g\|_{L^2([0,T];V)} + C_{\tilde{\varepsilon}} \|J_\varepsilon g - g\|_{L^2([0,T];V')}.\end{aligned}$$

We can now choose $\varepsilon > 0$ such that

$$\sup_n C_{\tilde{\varepsilon}} \|J_\varepsilon g_n - g_n\|_{L^2([0,T];V')} < \tilde{\varepsilon}$$

so that

$$\limsup_{n \to \infty} \|g_n - g\|_{L^2([0,T];H)} \le 2\tilde{\varepsilon} \sup_n \|g_n - J_\varepsilon g_n\|_{L^2([0,T];V)} + 2\tilde{\varepsilon}$$

which implies the assertion, since $\tilde{\varepsilon}$ was arbitrary.